

is estimated to be more than fifteen years.

As efficiency of polycrystalline solar cells is reasonable, these have a long life and low price. These solar cells are most suited for dual-purpose applications, provided there are no space constraints.

Morphology of PV device

Light is made of tiny particles called photons. A beam of sunlight is like a bright yellow fire hose shooting trillions upon trillions of photons in its way. If a solar cell is kept in its path, it catches energetic photons. These photons strike electrons in the silicon which are ejected, resulting in the formation of holes—vacancies left behind by escaping electrons.

If this happens in an electric field, the field moves the electrons to n-type layer and holes to p-type layer. When n-type and p-type layers are connected with a metallic wire,

electrons travel from n-type layer to p-type layer by crossing the depletion zone and then going through an external wire at the back of n-type layer, creating a flow of electricity (Fig. 3).

A solar cell consists of a layer of p-type silicon placed next to a layer of n-type silicon. In n-type layer there are excess electrons; and in p-type layer there are excess positively-charged holes. The n-type silicon is made by including atoms that have one more electron in their outer level as compared to silicon, such as phosphorus.

Phosphorus has five electrons in its outer level while silicon has four electrons. It bonds with its silicon neighbour atoms, but one electron is not involved in bonding. Instead, it is free to move inside the silicon structure. The p-type silicon is produced by adding atoms, such as boron or gallium, that have one less electron in their outer energy level than silicon. Because boron has one less electron than is required to form bonds with the surrounding silicon atoms, an electron vacancy or hole is created.

When these two materials are placed side by side inside a solar cell, n-type silicon's spare electrons jump over to fill the gaps in p-type silicon. This means, n-type silicon becomes positively-charged and p-type silicon becomes negatively-charged, creating an electric field across the cell. Near the junction of the two layers, electrons on one side of the junction (n-type layer) move into the holes on the other side of the junction (p-type layer). This creates an area around the junction, called depletion zone, in which electrons fill the holes, as shown in Fig. 4.

Because silicon is a semiconductor, it can act like an insulator, maintaining imbalance. As photons smash electrons off silicon atoms, this field drives them along in an orderly manner, providing electric current in the external circuit.

Anatomy of a polycrystalline solar cell

Fig. 5 shows a typical cross-section of a solar cell. An optimum silicon solar cell with light trapping and very good surface passivation is about hundred micrometres thick. However, thickness between 200 to 500 micrometres

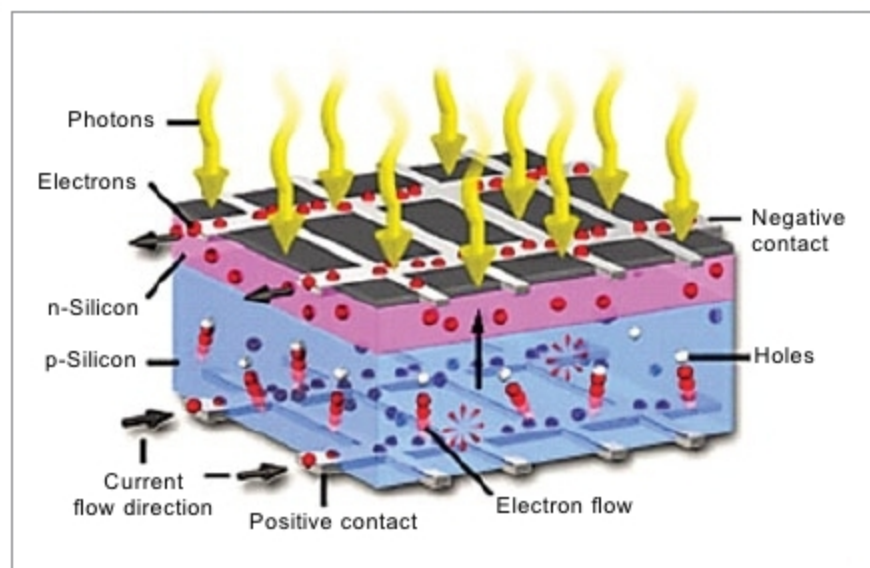


Fig. 3: Mechanism of current flow in photovoltaic cell

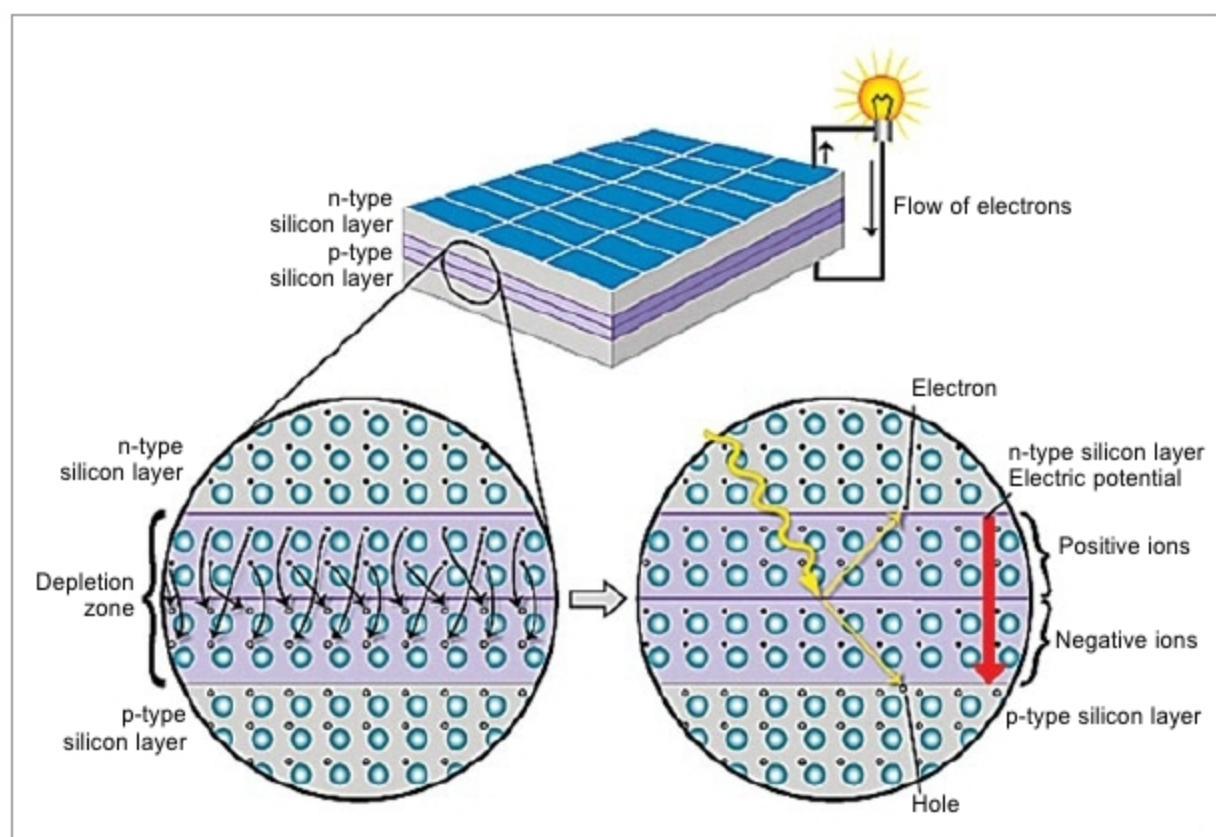


Fig. 4: Schematics of depletion zone in solar cells

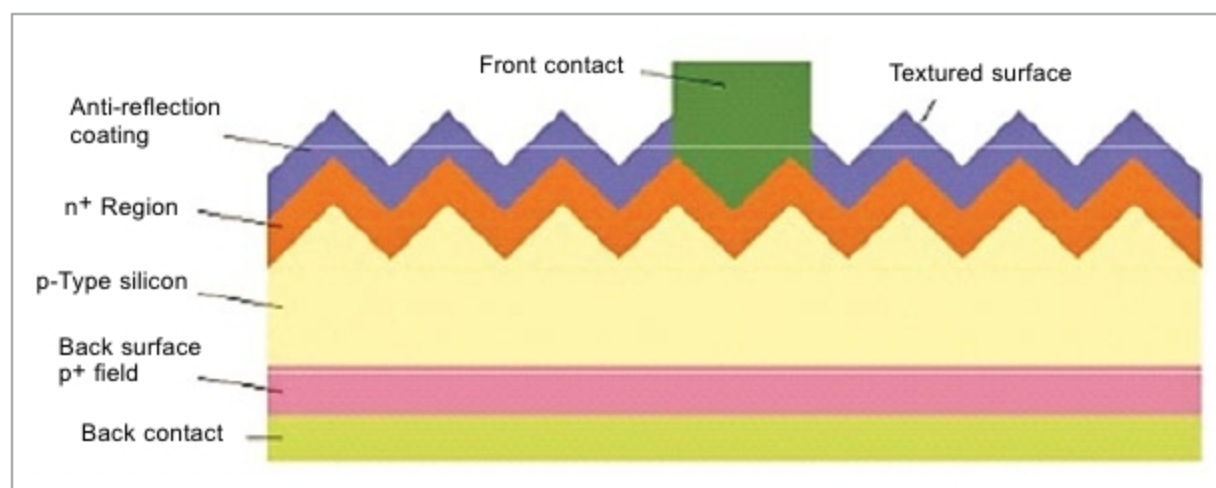


Fig. 5: Anatomy of polycrystalline solar cell